

1. Workload

2. ISA & ABI

3. Compute Org

4. Memory & NoC

5. Power Intent

6. Compiler/RT

7. Physical Node

8. Verification



CHAPTER STACK SCOPE — FULL STACK ACTIVE (STUDIES 1-2)

Running the Loop: End-to-End Case Studies

1. Study Setup

- Lighthouse Cache Study
- Systolic Array Study
- Pre-Registered Hypotheses
- Bound Parameter Baseline

2. Loop Execution

- Candidate Sweeps
- Simulator Run Logging
- Surrogate vs Simulator
- Resource Cost Tracking

3. Empirical Results

- Cache Study: Stopped
- Array Study: Tied Result
- Provenance Audit Card
- Cost-Benefit Accounting

4. Retained Verdict

- Baseline Retained
- Lesson Learned Record
- Design-Loop Card Signoff
- Evidence Handoff

Case Study Lessons & Retrospective Loop